

# Rates of Change and Rectilinear Motion

## ANSWER KEY



### Position, velocity, and acceleration

- 1 A particle moves along a line with position  $s(t) = t^3 - 6t^2 + 9t$  (metres, seconds).
    - a Find the velocity function  $v(t)$ .  $v(t) = 3t^2 - 12t + 9$ .
    - b Find the acceleration function  $a(t)$ .  $a(t) = 6t - 12$ .
    - c Find  $v(2)$  and  $a(2)$ .  $v(2) = 12 - 24 + 9 = -3$  m/s;  $a(2) = 12 - 12 = 0$  m/s<sup>2</sup>.
  - 2 For the particle above, find all times  $t \geq 0$  when the particle is at rest ( $v(t) = 0$ ).  
 $3t^2 - 12t + 9 = 0 \Rightarrow t^2 - 4t + 3 = 0 \Rightarrow (t - 1)(t - 3) = 0$ :  $t = 1$  and  $t = 3$ .
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### Speeding up and slowing down

- 3 Using the particle from above, determine whether the particle is speeding up or slowing down at  $t = 2$  (compare the signs of  $v(2)$  and  $a(2)$ ).  
 $v(2) = -3$  (negative) and  $a(2) = 0$ . Since  $a(2) = 0$ , the particle is neither speeding up nor slowing down at that instant — it is at a local extremum of speed.
  - 4 Determine whether the particle is speeding up or slowing down at  $t = 0$ .  
 $v(0) = 9 > 0$ ;  $a(0) = -12 < 0$ . Since  $v$  and  $a$  have opposite signs, the particle is slowing down at  $t = 0$ .
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### Total distance vs. displacement

- 5 For the particle above, find its displacement over  $[0, 3]$ .  
Displacement =  $s(3) - s(0) = (27 - 54 + 27) - 0 = 0$  m.
  - 6 Using the fact that the particle changes direction at  $t = 1$  (found earlier), find the total distance traveled over  $[0, 3]$ .  
 $s(0) = 0$ ,  $s(1) = 1 - 6 + 9 = 4$ ,  $s(3) = 0$ . Distance  
 $= |s(1) - s(0)| + |s(3) - s(1)| = |4 - 0| + |0 - 4| = 4 + 4 = 8$  m.
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### Related rates

- 7 A spherical balloon is inflated so its radius grows at 2 cm/s. Find the rate of change of its volume when  $r = 6$  cm, using  $V = \frac{4}{3}\pi r^3$ .  
 $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(36)(2) = 288\pi \approx 904.8$  cm<sup>3</sup>/s.

- 8 A 13 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top of the ladder is sliding down when the base is 5 m from the wall.
- $x^2 + y^2 = 169$ :  $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$ . At  $x = 5$ :  $y = 12$ .  $5(3) + 12\frac{dy}{dt} = 0$ , so  $\frac{dy}{dt} = -\frac{15}{12} = -1.25$  m/s (sliding down at 1.25 m/s).
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### Motion from a velocity graph

- 9 A particle's velocity is  $v(t) = t^2 - 4t$  (m/s) for  $t \geq 0$ . Find the times when the particle changes direction.
- $v(t) = t(t - 4) = 0$ :  $t = 0$  and  $t = 4$ . The particle changes direction at  $t = 4$  (velocity changes sign from negative to positive);  $t = 0$  is just the start.
- 10 For the same particle, find the acceleration at  $t = 1$  and state whether the particle is speeding up or slowing down at that instant.
- $a(t) = 2t - 4$ ;  $a(1) = -2$ .  $v(1) = 1 - 4 = -3$  (negative). Since  $v(1)$  and  $a(1)$  have the same sign (both negative), the particle is speeding up at  $t = 1$ .